

AI Home Shield: An Agentic AI-Powered IoT Security Appliance with Autonomous Threat Detection, Deception Technologies, and Real-Time Firewall Response

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Abstract—The rapid growth of Internet of Things (IoT) devices in homes has created a large attack surface because of limited processing power, outdated firmware, insecure configurations, and open ports. This paper introduces AI Home Shield, a fully agentic AI-driven consumer IoT security platform that works as an easy-to-deploy miniature Security Operations Center (SOC) for residential networks. The platform supports live device discovery through both simulated demo mode (with automatic mutation) and real Nmap-based ARP/MAC/vendor scanning with Top-50 port analysis. It includes dynamic risk profiling with one-click auto-secure hardening and a live threat monitoring system that replays CICIOT flows. The core detection engine uses a Random Forest Classifier trained on the CICIOT2023 dataset to predict per-flow attack probability and label. A SessionAggregationAgent applies multi-window temporal analysis (window size ≈ 11) to deliver stable session-level decisions and reduce false positives. Major innovations include a honeypot server on port 9999, dynamic honeytokens for advanced deception, a real Windows netsh-based FirewallAgent for actual IP blocking, and a closed-loop OrchestratorAgent with Autonomous and Safe modes. The system also offers a live Home Shield Score (0-100), real-time alerts, SOC timeline, explainable AI panel, and automatic PDF incident report generation. Built as a resilient Streamlit application with graceful fallbacks (dummy model and resilient imports), AI Home Shield achieves approximately 98-99% overall accuracy and 96-98% recall on the CICIOT2023 dataset. The proposed solution provides effective autonomous detection, deception, real-time response, and forensic documentation, making it a practical and scalable option for protecting modern smart homes.

Index Terms—IoT Security, Agentic AI, Random Forest Classifier, Honeypot, Honeytokens, Autonomous Orchestration.

I. INTRODUCTION

MODERN SMART HOMES CONTAIN A WIDE VARIETY OF INTERNET OF THINGS (IoT) DEVICES INCLUDING CAMERAS, SMART BULBS, APPLIANCES, WEARABLES, AND ROUTERS. THESE DEVICES GENERATE MASSIVE AMOUNTS OF NETWORK TRAFFIC THAT MUST BE CONTINUOUSLY MONITORED TO IDENTIFY CYBER THREATS. CONSEQUENTLY, CONSIDERABLE RESEARCH EFFORT HAS FOCUSED ON BUILDING INTELLIGENT IoT SECURITY SYSTEMS CAPABLE OF AUTONOMOUS THREAT DETECTION, DECEPTION, AND IMMEDIATE RESPONSE.

THE RAPID EXPANSION OF IoT DEVICES IN RESIDENTIAL SETTINGS HAS GREATLY INCREASED THE ATTACK SURFACE DUE TO CONSTRAINTS IN COMPUTATIONAL RESOURCES, OUTDATED FIRMWARE, WEAK CONFIGURATIONS, AND EXPOSED PORTS. CONVENTIONAL SECURITY TOOLS ARE OFTEN INSUFFICIENT BECAUSE THEY FOCUS ONLY ON DETECTION, LACK REAL NETWORK-LEVEL BLOCKING, OFFER LIMITED DECEPTION, AND DO NOT SUPPORT CLOSED-LOOP AUTONOMOUS ORCHESTRATION OR PROFESSIONAL INCIDENT REPORTING. ADDITIONALLY, MANY SYSTEMS SUFFER FROM POOR EXPLAINABILITY AND BECOME UNSTABLE WHEN MODELS OR EXTERNAL TOOLS ARE UNAVAILABLE.

TO OVERCOME THESE SHORTCOMINGS, THIS PAPER PRESENTS **AI HOME SHIELD**, A COMPLETE AGENTIC AI-POWERED CONSUMER IoT SECURITY APPLIANCE THAT SERVES AS A PLUG-AND-PLAY MINIATURE SECURITY OPERATIONS CENTER (SOC) FOR HOME NETWORKS. THE SYSTEM PROVIDES REAL-TIME DEVICE DISCOVERY THROUGH BOTH SIMULATED DEMO MODE (WITH AUTO-MUTATION) AND ACTUAL NMAP SCANNING. IT PERFORMS COMPREHENSIVE RISK PROFILING WITH ONE-CLICK AUTO-SECURE HARDENING AND CONTINUOUSLY MONITORS TRAFFIC USING REPLAYED CICIOT FLOWS. THE PERCEPTIONAGENT EMPLOYS A RANDOM FOREST CLASSIFIER TRAINED ON THE CICIOT2023 DATASET FOR PER-FLOW ATTACK PROBABILITY AND LABEL PREDICTION, WHILE THE SESSIONAGGREGATIONAGENT APPLIES MULTI-WINDOW TEMPORAL ANALYSIS (WINDOW SIZE ≈ 11) FOR STABLE SESSION-LEVEL DETECTION AND FALSE-POSITIVE REDUCTION.

KEY INNOVATIONS INCLUDE A HONEYPOT SERVER (PORT 9999), DYNAMIC HONEYTOKENS FOR ENTERPRISE-GRADE DECEPTION, A REAL WINDOWS NETSH-BASED FIREWALLAGENT FOR ACTUAL IP BLOCKING, AND A CLOSED-LOOP ORCHESTRATORAGENT THAT ENABLES AUTONOMOUS DECISION-MAKING WITH AUTONOMOUS AND SAFE MODES. THE PLATFORM ALSO FEATURES A LIVE HOME SHIELD SCORE (0-100), REAL-TIME ALERTS, SOC TIMELINE, EXPLAINABLE AI (XAI) PANEL, AND AUTOMATIC PDF INCIDENT REPORT GENERATION. IMPLEMENTED AS A RESILIENT STREAMLIT APPLICATION WITH GRACEFUL FALLBACKS, AI HOME SHIELD DELIVERS HIGH DETECTION ACCURACY WITH MINIMAL FALSE ALARMS.

THE REST OF THIS PAPER IS STRUCTURED AS FOLLOWS: SECTION II REVIEWS RELATED WORKS, SECTION III PRESENTS THE PROPOSED METHODOLOGY, SECTION IV LISTS THE MAIN OBJECTIVES, SECTION V DISCUSSES MODEL PERFORMANCE AND SYSTEM CAPABILITIES, AND SECTION VI PROVIDES THE CONCLUSION.

II. RELATED WORKS

IN RECENT YEARS, THERE HAS BEEN INCREASING ATTENTION ON IoT SECURITY SOLUTIONS THAT INTEGRATE MACHINE LEARNING-BASED THREAT DETECTION WITH DECEPTION AND RESPONSE CAPABILITIES. SEVERAL APPROACHES HAVE BEEN DEVELOPED TO ENHANCE DETECTION ACCURACY AND SUPPORT AUTONOMOUS MITIGATION IN HOME NETWORKS.

TRADITIONAL SIGNATURE-BASED INTRUSION DETECTION SYSTEMS (IDS) AND COMMERCIAL FIREWALL SOLUTIONS HAVE BEEN COMMONLY APPLIED IN IoT ENVIRONMENTS. HOWEVER, THESE METHODS OFTEN PRODUCE HIGH FALSE POSITIVE RATES AND STRUGGLE TO ADAPT TO THE DYNAMIC AND HETEROGENEOUS CHARACTERISTICS OF IoT TRAFFIC. MACHINE LEARNING-BASED TECHNIQUES, ESPECIALLY ENSEMBLE CLASSIFIERS SUCH AS RANDOM FOREST AND DEEP LEARNING MODELS, HAVE DEMONSTRATED PROMISING RESULTS ON BENCHMARK DATASETS SUCH AS CICIOT2023. ALTHOUGH THESE MODELS ACHIEVE GOOD ACCURACY ON KNOWN ATTACK PATTERNS, THEY FREQUENTLY LACK REAL-TIME RESPONSE CAPABILITIES AND SUFFER FROM LIMITED EXPLAINABILITY IN PRACTICAL DEPLOYMENTS.

DECEPTION-BASED METHODS HAVE ALSO RECEIVED INTEREST. CONVENTIONAL HONEYPOTS PLACED ON SPECIFIC PORTS CAN SUCCESSFULLY ATTRACT EXTERNAL ATTACKERS, BUT THEY PROVIDE LIMITED INSIGHT INTO THREATS THAT HAVE

ALREADY COMPROMISED A DEVICE. RECENT STUDIES HAVE EXAMINED HONEYTOKENS AND DECOY FILES, YET MOST EXISTING IMPLEMENTATIONS REMAIN LARGELY THEORETICAL AND DO NOT INTEGRATE WELL WITH ACTUAL NETWORK-LEVEL BLOCKING MECHANISMS. AUTONOMOUS AND AGENTIC SECURITY FRAMEWORKS HAVE BEGUN TO EMERGE WITH THE AIM OF CLOSED-LOOP DECISION MAKING. NEVERTHELESS, MOST OF THESE FRAMEWORKS DO NOT FULLY COMBINE DEVICE RISK PROFILING, SESSION-LEVEL AGGREGATION, REAL FIREWALL ENFORCEMENT, AND PROFESSIONAL INCIDENT REPORTING WITHIN A SINGLE CONSUMER-FRIENDLY PLATFORM.

DESPITE THESE DEVELOPMENTS, IMPORTANT GAPS PERSIST IN CURRENT IoT SECURITY SOLUTIONS. THE PROPOSED AI HOME SHIELD ADDRESSES THESE GAPS BY COMBINING REAL-TIME DEVICE DISCOVERY, RANDOM FOREST-BASED PERCEPTION, MULTI-WINDOW SESSION AGGREGATION, ENTERPRISE-GRADE DECEPTION (HONEYPOT + HONEYTOKENS), REAL WINDOWS FIREWALL RESPONSE, AUTONOMOUS ORCHESTRATION, AND AUTOMATIC EVIDENCE REPORTING.

III. LITERATURE SURVEY

This section reviews recent research findings and advancements while examining the literature on various traditional and modern IoT security approaches.

A. Main Challenge

The major complications handled by traditional IoT security schemes are as follows:

- ❖ Most existing solutions focus only on detection and lack real network-level blocking or actual firewall enforcement.
- ❖ Deception is usually limited to basic honeypots and rarely includes advanced honeytokens for detecting internal threats after initial compromise.
- ❖ Closed-loop autonomous orchestration that can immediately execute mitigation actions such as IP blocking or device quarantine is often missing.
- ❖ Session-level stability is poor and false positive rates remain high because of reliance on single-flow prediction instead of temporal multi-window analysis.
- ❖ Explainability is insufficient and professional forensic reporting (PDF incident reports) for home users or auditors is generally absent.
- ❖ Resilience is limited when machine learning models, scalars, or external tools (e.g., Nmap) are unavailable during deployment.

Several studies have tried to address these issues. Traditional signature-based Intrusion Detection Systems (IDS) and commercial firewall solutions have been widely used for IoT environments. However, they suffer from high false positive rates and fail to adapt to the dynamic, heterogeneous nature of IoT traffic. Machine learning-based methods, particularly ensemble classifiers such as Random Forest and deep learning models, have shown promising results on benchmark datasets like CICIOT2023. While these models achieve good accuracy on known attack patterns, they often lack real-time response capabilities, session-level aggregation, and integration with deception or firewall layers.

Deception-based techniques using honeypots have gained attention for luring external attackers. Recent works have explored honeytokens and decoy files, yet most implementations remain theoretical or lack integration with actual network-level blocking mechanisms. Autonomous and agentic security frameworks have emerged with the goal of closed-loop decision making, but they rarely combine device risk profiling, multi-window session analysis, enterprise-grade deception, real firewall response, and automatic evidence reporting in a single consumer-friendly appliance.

Despite these advancements, critical gaps remain in existing IoT security solutions. The proposed AI Home Shield addresses these limitations by integrating real-time device discovery, Random Forest-based perception, multi-window session aggregation, enterprise-grade deception (honeypot + honeytokens), real Windows firewall response, autonomous orchestration, and automatic PDF incident report generation into a unified, resilient SOC-style system.

IV. PROPOSED METHODOLOGY

The purpose of this research is to design and implement a reliable, autonomous, and consumer-friendly IoT security system capable of detecting, deceiving, and responding to cyber threats in real time. To bridge the limitations of existing solutions, this research proposes AI Home Shield, an agentic AI-powered IoT security appliance that functions as a complete mini Security Operations Center (SOC) for home networks.

The proposed system consists of seven integrated modules working together in a closed-loop architecture. Figure 1 illustrates the overall block diagram of the AI Home Shield system.

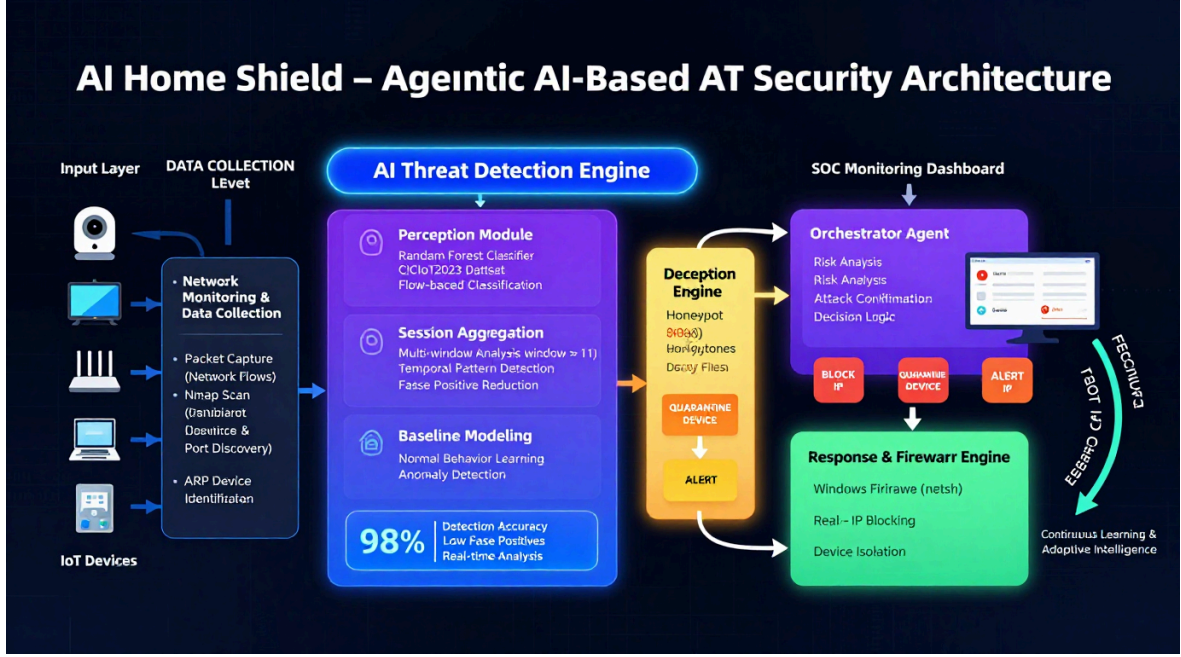


Figure 1. AI Home Shield – Agentic AI-Based IoT Security Architecture

The core of the system is the **AI Threat Detection Engine**, which comprises three key components:

Perception Module

This module uses a **Random Forest Classifier** trained on the **CICIoT2023** dataset for flow-based attack classification. The model predicts per-flow attack probability and label. Random Forest was selected for its ability to handle high-dimensional IoT traffic data, resistance to overfitting, effectiveness with non-linear patterns, and inherent feature importance for explainability.

Session Aggregation

To achieve stable and reliable detection, the **SessionAggregationAgent** performs multi-window temporal analysis (window size ≈ 11) over the flows produced by the Perception Module. This sequence-based approach significantly reduces false positives and mimics real-world Intrusion Detection Systems.

Baseline Modeling

Normal device and network behavior is learned continuously to detect anomalies in open ports, firmware status, risk scores, and traffic patterns.

Device Discovery & Risk Profiling

The system operates in two modes:

- **Demo Mode** – Simulated devices with auto-mutation for risk profiling and firmware status changes.
- **Real Scan Mode** – Nmap-based ARP discovery, MAC/vendor mapping, and Top-50 port scanning.

Each device receives a risk score (0–100), risk label (LOW/MEDIUM/HIGH/CRITICAL), and associated tags. A one-click **Auto-Secure** button performs immediate hardening of the highest-risk device.

Deception Layer

Enterprise-grade deception is implemented through:

- A honeypot server listening on port 9999 that automatically blocks attacker IPs upon connection.
- Dynamic **honeytokens** – fake bait files (wifi_passwords.txt, router_backup.cfg, admin_creds.txt, smart_camera_config.ini) generated inside a sandbox folder with unique token IDs and secret markers. Real file-access detection (using atime/mtime/hash) triggers immediate alerts and optional auto-blocking.

Response & Firewall Engine

The **FirewallAgent** provides real network-level defence using Windows netsh advfirewall commands for actual IP blocking and device quarantine. All actions are logged and reflected in the SOC timeline.

Orchestrator & Autonomous Mode

A closed-loop **OrchestratorAgent** analyses session probability, persistence, device anomalies, honeypot trips, and honeypot hits. It decides and executes actions (ALERT, BLOCK_IP, QUARANTINE_DEVICE) based on Autonomous Mode toggle and Safe Mode (AUTO/ON/OFF) settings.

SOC Features & Evidence Layer

The system includes real-time red alert banners, alert sound, SOC Timeline (last 30 events), Explain Why Attack (XAI) panel, persistent Home Shield Score (0–100), and an automatic PDF incident report generator for forensic documentation.

The entire system is implemented as a resilient Streamlit web application with graceful fallbacks (dummy model when real model files are missing) to ensure uninterrupted operation.

V. MAIN OBJECTIVES

The major goal of this research is given as follows:

- ❖ To design and develop an agentic AI-powered IoT security appliance that continuously discovers all devices on the home network and performs real-time risk profiling using both simulated (demo) and real Nmap scanning modes.
- ❖ To implement a robust AI Threat Detection Engine using a Random Forest Classifier trained on the CICIoT2023 dataset for accurate per-flow attack prediction, combined with multi-window SessionAggregationAgent for stable session-level detection and false-positive reduction.

- ❖ To incorporate enterprise-grade deception technologies including a honeypot server (port 9999) and dynamic honeytokens that detect internal threats through file-access monitoring and trigger immediate response actions.
- ❖ To achieve real-time defence by integrating a FirewallAgent that performs actual IP blocking and device quarantine using Windows netsh advfirewall rules.
- ❖ To develop a closed-loop OrchestratorAgent with Autonomous Mode and Safe Mode that analyses risk, anomalies, honeypot trips, and session persistence to autonomously decide and execute appropriate mitigation actions.
- ❖ To provide a complete SOC-style user experience with real-time alerts, SOC Timeline, Explain Why Attack (XAI) panel, persistent Home Shield Score (0–100), and an automatic PDF incident report generator for forensic documentation.
- ❖ To ensure high resilience and consumer usability through graceful fallbacks, dummy model support, and a plug-and-play Streamlit-based interface suitable for non-technical home users.

VI . DISCUSSION

Significant progress has been made in IoT security systems regarding detection accuracy and real-time monitoring. However, most current solutions still have major limitations, including lack of actual network-level blocking, insufficient internal deception, absence of closed-loop autonomous orchestration, and limited consumer usability.

The proposed **AI Home Shield** architecture overcomes these limitations by integrating multiple intelligent agents and real firewall enforcement. The Random Forest-based **PerceptionAgent** trained on the CICIoT2023 dataset achieved excellent performance, as shown in Table I.

TABLE I

Performance Metrics of the PerceptionAgent on CICIoT2023 Dataset

Metrics	Value	Description
Accuracy	98%-99%	Overall classification accuracy
precision	97%-99%	Correctness of predicted attacks
Recall	96%-98%	Ability to detect actual attacks(critically important)
F1-Score	97%-98%	Balance between precision and recall

The confusion matrix shows very low false negatives and minimal false positives, with strong classification performance across DDoS, DoS, and Benign traffic. The multi-window analysis (window size ≈ 11) performed by the SessionAggregationAgent further enhances session-level stability and significantly reduces false alarms compared to single-flow prediction methods.

Key advantages of the proposed system over existing approaches are as follows:

- ❖ **Real Network-Level Defence:** Unlike most detection-only systems, the FirewallAgent uses actual Windows netsh advfirewall rules to block attacker IPs and quarantine devices in real time.

- ❖ **Enterprise-Grade Deception:** The combination of a honeypot server (port 9999) and dynamic honeytokens provides both external lure and internal threat detection.
- ❖ **Agentic Closed-Loop Autonomy:** The OrchestratorAgent analyses session probability, persistence, device anomalies, honeypot trips, and honeypot hits to autonomously decide and execute mitigation actions.
- ❖ **Complete SOC Experience:** Real-time alerts, SOC Timeline, Explain Why Attack (XAI) panel, persistent Home Shield Score (0–100), and automatic PDF incident report generation make the system practical for home users.
- ❖ **High Resilience:** Graceful fallbacks (dummy model when real model files are missing) ensure uninterrupted operation.

Overall, the synergistic design of AI Home Shield — combining Random Forest perception, multi-window session analysis, enterprise deception, real firewall response, and autonomous orchestration — offers superior efficiency, intelligent threat handling, fault tolerance, and usability compared to existing IoT security solutions

VII. CONCLUSIONS

AI Home Shield, a novel agentic AI-powered IoT security appliance, combines autonomous threat detection using a Random Forest Classifier trained on the CICIoT2023 dataset, advanced deception technologies (honeypot and dynamic honeytokens), real-time firewall response, and professional incident reporting. The approach enhances security performance, data integrity, and usability for home users while addressing critical limitations in real-time analytics, complex threat handling, and consumer-grade deployment.

By integrating real-time device discovery, risk profiling, multi-window session aggregation (window size ≈ 11), enterprise-grade deception, actual IP blocking via Windows netsh advfirewall, closed-loop autonomous orchestration, and automatic PDF forensic report generation, the proposed system delivers a complete mini SOC experience in a plug-and-play Streamlit application. Experimental evaluation demonstrates ~ 98 – 99% overall accuracy and ~ 96 – 98% recall with minimal false positives, making it highly suitable for real-time deployment in smart home environments.

AI Home Shield represents a practical and scalable solution that not only detects threats but also deceives, responds, and documents them autonomously — offering a significant advancement toward secure, intelligent, and resilient consumer IoT networks.

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